
The complex image method for urban scenarios

Seminararbeit/ Seminar Work

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1 Introduction

Sound waves propagate from a source into an environment and are thereby interacting with various objects. In outdoor and urban environments, sound from cars, construction sites, or machines is reflected by the ground or buildings, scattered by trees, or diffracted around noise barriers. In addition, medium properties lead to phenomena such as refraction and attenuation. Understanding and modeling these effects is not only important for plausible auralization, but also because outdoor sound propagation touches nonacoustical subjects such as sociology, economics, regulations, and standards [1, 2]. At the same time, this is challenging due to the need for approximations based on object dimensions relative to acoustic wavelengths, which range from meters to centimeters for the audible sound. Consequently, sound propagation models are valid only within specific frequency ranges [1]. A frequently used assumption is that sound from sources is considered as a plane wave. In some cases, however, this may not hold true. Measurements outdoors showed surprising effects which cannot be explained by a simple plane wave reflection, and are therefore referred to as *ground effects* [3, 4]. This work will discuss the *Complex Image Method*, which considers spherical wave propagation in order to model the observed effects due to the ground [5, 4].

After a brief overview of the acoustic fundamentals of sound propagation and the commonly used *Simple Image Method* in Chapter 2, the observed features due to the ground will be discussed referring to measurements from the IHTA-Park in Chapter 3. Chapter 4 presents the idea of the derivation of the *Complex Image Method* with a spherical wave reflection factor and introduces impedance models for characterizing ground surfaces. After a discussion of previously proposed methods including simulations and a comparison of available methods in Chapter 5, Chapter 6 concludes with a summary and outlook.

2 Fundamentals

This chapter introduces the acoustic fundamentals and reflection as a fundamental effect of outdoor sound propagation, which leads to the Simple Image Method.

2.1 Basic Concepts of Wave Propagation

Sound usually originates from sources. If a sound source is small compared to the distances being considered, it is valid to assume these sources as point sources with waves propagating in three dimensions evenly. The complex sound pressure of a point source in a free field is given by

$$\underline{p}(r, k) = \frac{j\omega\rho_0\hat{Q}}{4\pi r} e^{jkr}. \quad (2.1)$$

$\hat{Q}$ is the volume velocity, $\omega = 2\pi f$ the angular frequency, where f is the frequency, $k = \frac{\omega}{c}$ is the wave number with c being the speed of sound, ρ_0 the density of air, r is the radial distance from the source and j is the imaginary unit. Note that in this work the phase convention e^{+jkr} is used for an outgoing wave to be consistent with the literature [4, 6, 5]. The same applies to plane waves.

In an ideal free field, a point source radiates spherical wavefronts and thus has for reasons of energy conservation a -6 dB reduction in sound pressure level for each doubling the distance from the source. If a source is located in the far field, it can be assumed that the wavefronts are flat and parallel, which simplifies many calculations. However, in real-world outdoor scenarios, several fundamental effects influence sound propagation:

Reflection occurs when sound waves hit surfaces such as buildings, the ground, or other obstacles, which changes the sound direction, leads to echoes, and potentially amplifies or cancels the direct sound due to constructive respectively destructive interference [1].

Scattering happens when sound waves encounter small obstacles or rough or structured surfaces in orders of magnitude as the wavelength, causing diffusing, spreading in different directions, and reducing the coherence of the sound wave [1].

Diffraction describes the effect that sound waves bend around large obstacles and through small openings, en-

abling the sound to reach areas that are not in the direct line of sight of the source, which is an important effect in environments with e.g. buildings or noise barriers at highways [1]. *Refraction* is the bending of sound waves as they pass through layers of air with varying temperatures or densities. Near the ground temperature gradients can occur, which can change the speed and direction of the waves, bending them back towards the ground or up into the atmosphere [2]. *Attenuation* refers to the loss of sound energy as it propagates through the air, influenced by air absorption, humidity, and environmental conditions, which together reduce the sound pressure and intensity over distance [1].

As this work is focused on ground reflections, these will be introduced more closely in Section 2.2.

2.2 Reflection at an Impedance Plane

Let us first assume a plane wave hitting an infinitely large and smooth ground under an angle θ , which is measured between the ground normal and the propagation direction of the wave. According to Snell's law it is reflected as shown in Figure 2.1, which is commonly described as a *specular reflection* [1].

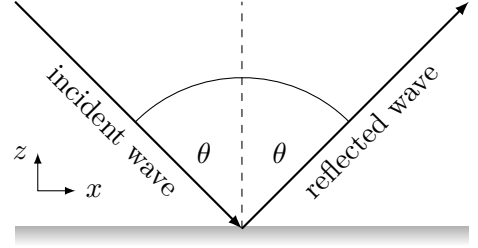


Figure 2.1: Specular Reflection

The sound pressure of the incident wave

$$\underline{p}_i = \hat{p} e^{j(\omega t + kz \cos \theta - kx \sin \theta)} \quad (2.2)$$

can be reduced and changed in phase depending on the characteristics of the ground, which results in the reflected wave

$$\underline{p}_r = \hat{p} \underline{R}(f, \theta) e^{j(\omega t - kz \cos \theta - kx \sin \theta)} \quad (2.3)$$

with the in general complex *plane wave reflection factor*

$$\underline{R}(f, \theta) = \frac{\underline{p}_r}{\underline{p}_i} = \frac{\underline{Z}(f) \cos(\theta) - Z_0}{\underline{Z}(f) \cos(\theta) + Z_0}, \quad (2.4)$$

which is connected with the ground impedance $\underline{Z}(f)$ and the characteristic impedance of air $Z_0 = \rho_0 c$. The ground impedance is in general frequency and also angle dependent, although the latter is not investigated further in this work, as for most cases the as-

sumption of a *local reaction* is sufficient. Another important parameter, which expresses how much of the incident sound energy is absorbed, is the absorption coefficient [1]

$$\alpha = 1 - |\underline{R}(f, \theta)|^2. \quad (2.5)$$

E.g. a rigid wall is characterized by $\underline{Z} = \infty$, $\underline{R}(f, \theta) = 1$ and $\alpha = 0$.

2.3 The Simple Image Method

To further analyze reflections and the resulting sound fields by superposition of the direct and reflected wave, e.g. a sound source above the ground or in front of a wall, the simple image method, which is also well known in the context of calculating electrostatic potentials, provides a useful approach, which avoids solving the wave equation [7]. As shown in Figure 2.2, a source is located at S above the ground. The total sound field at the receiver position R contains the direct sound traveling along r_0 and the reflected sound along r_1 . The latter can be obtained by mirroring the source S on the ground, which leads to an *Image Source* IS, and weighting it with the reflection factor. Using Eq. 2.1 for the point source simplified by setting $\frac{j\omega\rho_0\hat{Q}}{4\pi} = 1$, this gives the total resulting sound pressure at the receiver R [1]:

$$\underline{p}_R = \underline{p}_S + \underline{p}_{IS} = \frac{e^{ikr_0}}{r_0} + \frac{e^{ikr_1}}{r_1} \underline{R}(f, \theta_0) \quad (2.6)$$

Eq. 2.6, in fact, is an approximation as it is assuming a constant angle of incidence θ_0 and, thus, actually a plane wave, which is the central assumption for *geometrical acoustics* [1]. At the boundary, plane and spherical waves can be split into infinitely many elementary sources, also known as the Huygens principle. These elementary sources build up an interference field which produces a plane or spherical wave. This provides the basis for the simplicity using the plane wave approximation, described in Section 2.1, when reflection occurs. However, the *Simple Image Method* is even exact if $\underline{R} = \pm 1$.

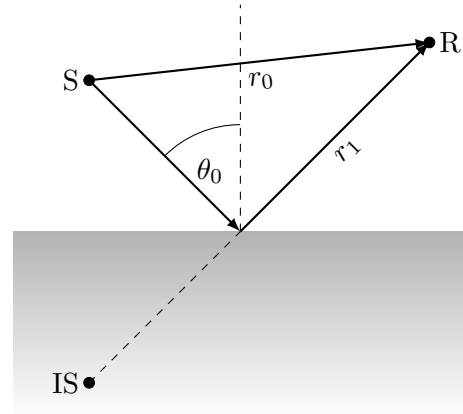


Figure 2.2: Simple Image Method

3 Phenomena in Outdoor Sound Propagation

In outdoor sound propagation of broadband sound over grassland, snow, or cultivated land near to the ground, two curious phenomena can be observed:

1. Frequencies between 250 Hz and 800 Hz suffer much greater attenuation than frequencies below 250 Hz or in the range of 800 Hz to 2 kHz,
2. a *negative attenuation* can be observed at frequencies below 250 Hz, meaning they are amplified relative to the expected levels if the ground surface were not present.

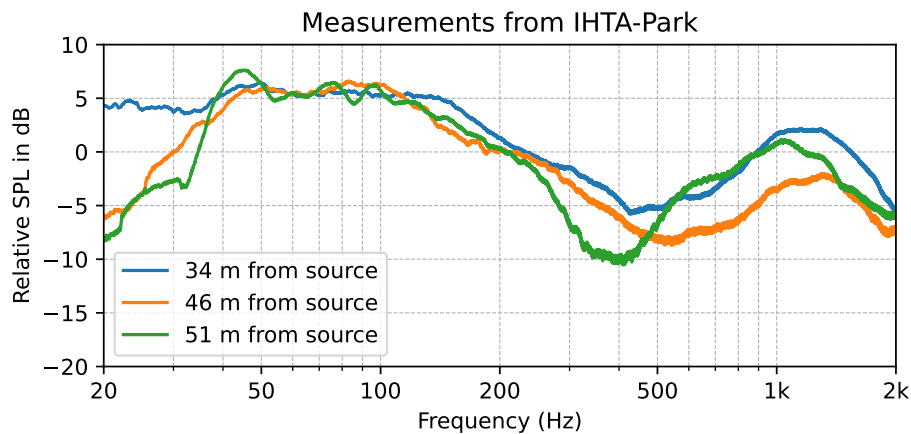


Figure 3.1: Relative sound pressure levels (SPL) in dB, measured at different distances from a point source 1.45m above the ground at a receiver height of 1.2m in the IHTA-Park, shown in Appendix A.1, Figure A.1.

These effects were first documented by Parkin and Scholes in 1965 [3, 4] and have also been present in recent measurements conducted in the IHTA Park, shown in Figure 3.1. This Chapter introduces and discusses the main effects and features in outdoor sound propagation due to the ground.

3.1 Proximity to the Ground

A typical scenario in outdoor sound propagation involves a sound source near the ground, a receiver a few meters above the ground at most, and a relatively large horizontal separation between the source and receiver compared to their altitudes.

In this setup, the grazing angle ξ of the specularly reflected sound ray with the surface is very small. As already discussed in Section 2.3, the specular reflection is not fully accurate when the wavelengths are similar to the dimensions of the configuration. The plane-wave reflection factor (Eq. 2.4) depends on the angle of incidence θ and the ratio of the acoustic impedances of air and the ground. For locally reacting surfaces, the reflection at a point is determined by the ground at that point and varies with the angle unless extreme conditions occur: infinitely hard ground, infinitely soft ground, or constant incidence angle. None of these conditions are practical as the angle of incidence is not constant due to the fact that the plane-wave approximation of the point source is not valid near the ground [2, 4], cf. Figure 3.2.

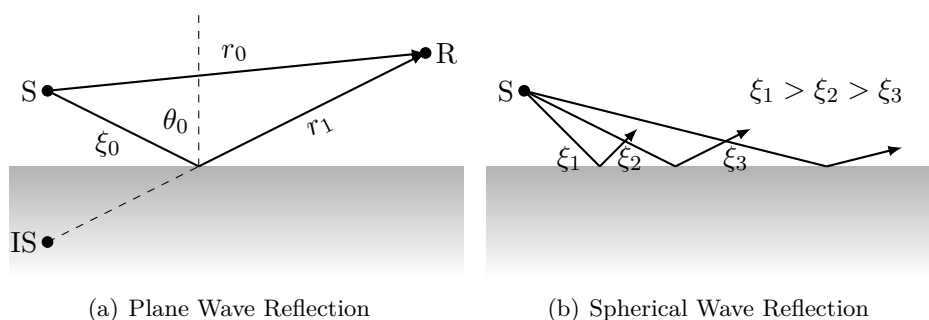


Figure 3.2: Schematic illustration of (a) the direct and reflected sound paths when source and receiver are near the ground, and (b) varying grazing angle of incidence with position along the surface when the incident wavefronts are not plane.

Thus, there cannot be a perfect image in the ground mirroring the source. Instead, there is a *fuzzy* image spread over a large area, strongest at the expected location but extending infinitely horizontally and downwards [2].

3.2 Interference

Due to the ground reflection, cf. Sections 2.2 and 3.1, the direct and reflected sound waves interfere with each other. This occurs when the direct sound wave from the source and the sound wave reflected off the ground meet at the receiver with a path difference.

Depending on their phase relation, these waves can either amplify or cancel each other. When the waves are in phase, i.e. peaks align with peaks, constructive interference occurs, and the sound pressure level increases. And the other way around, when the waves are out of phase, destructive interference occurs, reducing the sound pressure level. Over an acoustically rigid surface, $\underline{R} \rightarrow 1$, like asphalt or concrete, there is no phase change $\Delta\varphi$ on reflection at the ground. The observed sound pressure levels result entirely from the path length difference between the source and receiver. The frequencies for minima ($n = 0, 1, 2, \dots$) in the total pressure are in this case given by [4]

$$f_m = \frac{c(2n + 1)}{2(r_1 - r_0)}. \quad (3.1)$$

However, when looking at the measurements, cf. Figure 3.1, it is striking that the first interference minimum occurs at much lower frequencies and is broader than analytically expected for a rigid ground reflection with e.g. $f_{m,34m} = 1,4$ kHz for the 34 m distance and also does not explain the greater gain before the first minimum than after. This discrepancy suggests additional factors, such as ground waves and absorption influencing the observed sound field.

3.3 Acoustic Ground Waves

The study of sound propagation over grass-covered grounds conducted by Embleton [8] observed a rapid decrease in sound pressure levels at high frequencies when the receiver is near to the ground. He found that raising the receiver height recovers much of the lost high-frequency signal, but not totally, see Figure 3.3. This observation raises the question of why there is a significant loss in sound pressure at high frequencies in the absence of destructive interference [2]. When both the source and receiver are near the ground, which means $\cos(\theta) \rightarrow 0$ and according to Eq. 2.4, $\underline{R}(f, \theta) \rightarrow -1$, the reflected field should cancel the direct field, resulting in low sound-pressure levels at all frequencies. However, the presence of a strong signal below 800 Hz indicates the influence of a *ground wave*, an effect which cannot be fully explained by plane-wave reflection theories, cf. Sections 2.2 and 2.3.

In acoustical terms, the ground wave is the component of the reflected sound field that is not taken into account by the reflection coefficient for plane waves and occurs when the incident waves are not plane, as described in Section 3.1. Its origin is linked to the *fuzzy* part of the image source. This contributes additional reflected sound pressure, that varies with frequency and distance and is called *ground wave*.

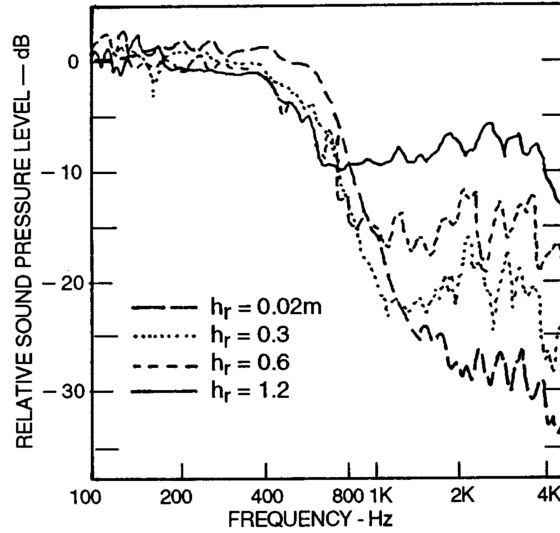


Figure 3.3: Relative sound pressure levels measured 5 m from a point source at an acoustically soft ground (grass) for different receiver heights h_r . Figure taken from [2].

3.4 Ground Impedance

Ground impedance significantly affects sound reflection, as shown in Eq. 2.4. Measurements of various grounds [9, 6] have demonstrated that the impedance depends heavily on the properties of the ground and is also frequency-dependent. The more porous a surface is, the more the first interference minimum shifts to lower frequencies and becomes broader, e.g. for snow or grass, in comparison to the behavior that is measured and expected for rigid grounds like asphalt. Studies have shown that effective flow resistivity is a suitable parameter for characterizing the ground. Models for determining ground impedance are introduced in Section 4.2.

4 Modeling Ground Effects

Since the simple plane wave reflection is not sufficient for describing the occurring effects for outdoor scenarios where source and receiver are near to the ground as outlined in the first chapters in detail, this chapter introduces the *Complex Image Method*, which contains a spherical reflection factor. Furthermore mathematical models for the ground impedance will be presented.

4.1 The Complex Image Method

In this section, Ingard's approach for the derivation of the *spherical reflection factor* [5] will be presented, which is closely related to Weyl's solution for the electromagnetic case [10]. The section does not aim to explain all calculations, which can be found in literature [5], but to comment on those steps, which are left out in literature but are very crucial to understand the derivation.

Let's consider a point source S at a distance h above an infinite plane, cf. Figure 4.1. The total sound pressure $\underline{p}_P$ at the receiver point $P = (x, y, z)$ is given by the sum of the direct wave $\underline{p}_S$ and the reflected wave from the image source Q $\underline{p}_Q$:

$$\underline{p}_P = \underline{p}_S + \underline{p}_Q. \quad (4.1)$$

Weyl's identity [10] can be used to express the spherical wave $\underline{p}_S$ in the half plane below the source as the linear combination of plane waves

$$\underline{p}_S = \frac{e^{jkr_1}}{r_1} = \frac{j}{2\pi} \int_A e^{jk(ax+by+c(h-z))} dA, \quad (4.2)$$

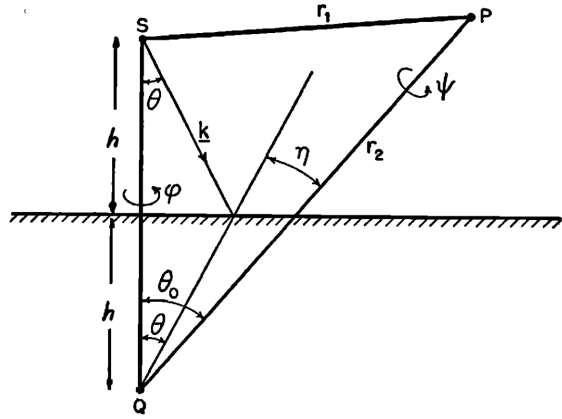


Figure 4.1: Reference Geometry (Ref. [5])

where $\mathbf{k} = (a, b, c)^T = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)^T$ with the azimuth angle ϕ measured from the x-axis in the direction of the y-axis and the polar angle θ from the z-axis downwards. $\mathbf{k} = (a, b, c)^T$ are the directional cosines, i.e. the direction of the plane wave, which is integrated, and r_1 is the distance from the source S to the receiver point P. $dA = \sin(\theta)d\theta d\phi$ is the the surface element of the unit sphere. ϕ is integrated from 0 to 2π , θ is integrated in the complex θ -plane from $\theta = 0$ to $j\infty$, which ensures the convergence of the integral [10] and will be explained in detail later. Analogically, the reflected wave can be written as a linear combination of plane waves weighted by the plane-wave reflection factor $\underline{R}(f, \theta)$ (Eq. 2.4):

$$\begin{aligned} p_Q &= \frac{j}{2\pi} \int_A \underline{R}(f, \theta) e^{jk(ax+by+c(h+z))} dA \\ &= \frac{j}{2\pi} \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\frac{\pi}{2}+j\infty} \underline{R}(f, \theta) e^{jkr_2 \cos(\theta)} \sin(\theta) d\theta d\phi. \end{aligned} \quad (4.3)$$

Rewriting the exponent $(ax + by + c(h + z)) = \mathbf{k}\mathbf{r}_Q = \|\mathbf{k}\| \|\mathbf{r}_Q\| \cos(\theta)$, where $\|\mathbf{k}\| = 1$ and $\|\mathbf{r}_Q\| = \|(x, y, (h + z))^T\| = r_2$ yields the equation above, which provides a mathematically correct solution for the problem, however, not the desired *spherical reflection factor* $\mathcal{Q}$. Cauchy's integral theorem states that two paths connecting the same points have the same path integral if the function is holomorphic between the two paths. That makes it possible to introduce a new coordinate system with Q as it's origin and with the direction QP as the new axis and η and ψ as the angle variables as shown in Figure 4.1 and, in particular, to choose the same integration path as used before:

$$p_Q = \frac{j}{2\pi} \int_{\psi=0}^{2\pi} \int_{\eta=0}^{\frac{\pi}{2}+j\infty} \underline{R}(f, \psi, \eta) e^{jkr_2 \cos(\eta)} \sin(\eta) d\eta d\psi. \quad (4.4)$$

The path of integration for η must be chosen in the direction of the steepest descent in the complex plane, which is illustrated in Appendix A.2 and leads to the substitution

$$\cos(\eta) = 1 + jt, \quad \text{and} \quad d\eta = \frac{-j}{\sin(\eta)} dt, \quad (4.5)$$

where t is a real parameter, and as jt varies, $\cos(\eta)$ traces out the illustrated path in the complex plane. The new upper limit is $t = \frac{\cos(\pi/2+j\infty)-1}{j} \rightarrow \infty$ by using Eulers identity. The point source can now be moved in front of the integral, the integration to ψ can be separated, see. Eq. 4.7, and the reflected wave can then be written as

$$p_Q = \frac{e^{jkr_2}}{r_2} r_2 j \int_{t=0}^{\infty} e^{-kr_2 t} \underline{R}(t)_{Av}(-j) dt = \frac{e^{jkr_2}}{r_2} \mathcal{Q}, \quad (4.6)$$

where

$$\underline{R}(t)_{Av} = \frac{1}{2\pi} \int_{\psi=0}^{2\pi} \underline{R}(t, \psi) d\psi, \quad (4.7)$$

and $\mathcal{Q}$ is the *spherical reflection factor*. $\underline{R}(t)_{Av}$ is the ensemble average of the plane-wave reflection factor. After a series of calculations detailed in [5], it is possible to separate the plane wave reflection factor (cf. Eq. 2.4)

$$\underline{R}_0 = \frac{\cos(\theta_0) - \underline{\beta}}{\cos(\theta_0) + \underline{\beta}}, \quad (4.8)$$

where $\underline{\beta} = Z_0/\underline{Z}(f)$ is the admittance. This makes sense since it is physically meaningful that $\mathcal{Q}$ will approach the plane wave reflection factor asymptotically. Since $kr_2 \int_{t=0}^{\infty} e^{-kr_2 t} dt = 1$, it follows from Eq. 4.6 and the solution of Eq. 4.7 that

$$\mathcal{Q} = \underline{R}_0 + (1 - \underline{R}_0)\mathcal{F}. \quad (4.9)$$

$\mathcal{F}$ follows out of a series of calculations detailed in [5]. The full set of needed equations summarizes to

$$\begin{aligned} \mathcal{Q} &= \underline{R}_0 + (1 - \underline{R}_0) [1 + j\sqrt{\pi\rho}e^{-\rho}\Phi(-j\sqrt{\rho})], \\ \rho &= \frac{jkr_2}{2} \frac{(\cos(\theta_0) + \underline{\beta})^2}{1 + \cos(\theta_0)\underline{\beta}}, \end{aligned} \quad (4.10)$$

where $\Phi(\cdot)$ is the complementary error function. The reflected sound field is finally a function of the distance and angle from the image source Q to the receiver point P and the ground impedance. The total pressure at the receiver P is given by

$$\underline{p}_P = \frac{e^{jkr_1}}{r_1} + \frac{e^{jkr_2}}{r_2} \mathcal{Q}. \quad (4.11)$$

4.2 Acoustic Characteristics of Porous Ground Surfaces

With the *Complex Image Method* it is possible to model the spherical reflection at an impedance plane in theory very precisely as we found an exact solution of the wave equation without any plane-wave assumption. However, having the big picture in mind, we must notice that this is heavily limited by the ground impedance, which has to match the reality as closely as possible to really benefit from the presented method. As it is very hard and often not possible to measure the ground impedance directly, many models can be found in the literature [11, 6, 9, 12].

Delany and Bazley Impedance Model [9] This single parameter semi-empirical model has been used widely to characterize outdoor ground surfaces. Based on impedance tube measurements on fibrous materials with porosity close to 1, the following expressions (Eq. 4.12) for the relative characteristic impedance $\underline{Z}_c$ and the propagation constant $\underline{k}_{ground}$ were obtained:

$$\begin{aligned}\frac{\underline{Z}_c}{Z_0} &= 1 + 9.08 \left(\frac{f}{\sigma} \right)^{-0.754} + j11.9 \left(\frac{f}{\sigma} \right)^{-0.732}, \\ \frac{\underline{k}_{ground}}{k} &= 1 + 10.8 \left(\frac{f}{\sigma} \right)^{-0.700} + j10.3 \left(\frac{f}{\sigma} \right)^{-0.595}.\end{aligned}\quad (4.12)$$

σ is the effective flow resistivity in rayls (cgs units), which means that 1 rayl equals $10^3 \frac{Pas}{m^2}$. Note that the phase convention for an outgoing spherical wave is as defined in Section 2.1. Otherwise, signs need to be changed, cf. [11]. The model is regarded as semi-empirical since f/σ is related to the ratio of the pore dimension to the thickness of the viscous boundary layer. This ratio has been used when deducing similar power law formulae for the acoustical properties of several types of porous materials. Values for the effective flow resistivity are depicted in Appendix A.3 in Table A.1.

For further models that take other parameters such as porosity, tortuosity, etc. into account in addition to the effective flow resistivity, please refer to the literature [12, 4, 11].

Hard-backed Layer Model [4] The normalized data for the characteristic impedance and propagation constant (presented above) find direct application when a sufficiently large thickness of material is used. For many applications, however, the performance of a relatively thin layer is required. The impedance $\underline{Z}_d$ of a rigidly-backed layer of thickness d can be calculated from

$$\frac{\underline{Z}_d}{Z_0} = \frac{\underline{Z}_c}{Z_0} \coth(-j\underline{k}_{ground}d). \quad (4.13)$$

Non Hard-backed Layer Model [4] Let's assume snow has fallen and covers the grassland. The assumption that the lower layer is hard-backed is not valid anymore. Thus, it might be necessary to consider a non-hard-backed-layer impedance model. This can be deduced from the characteristic impedances of the upper layer $\underline{Z}_1$ with the corresponding propagation constant $\underline{k}_1$ and the substrate $\underline{Z}_2$ using transmission line analysis

$$\frac{\underline{Z}_d}{Z_0} = \underline{Z}_1 \frac{\underline{Z}_2 - j\underline{Z}_1 \tan(\underline{k}_1)}{\underline{Z}_1 - j\underline{Z}_2 \tan(\underline{k}_1)}. \quad (4.14)$$

5 Case Studies and Comparison of Methods

This chapter discusses the characteristics introduced by the *Complex Image Method*. In addition, conventional and simplified approaches used in current implementations of image edge models [13] in outdoor auralization will be compared with the *Complex Image Method* and their potential benefits will be critically discussed.

Figure 5.1 shows the relative sound pressure level (SPL), which is the sound pressure at the receiver, i.e. direct and reflected sound field, normalized by the direct sound field at the receiver without any reflections. The graph illustrates the effect that a given type of ground will have on the reflected sound field above it for a source and receiver located $h_S = 0.3\text{m}$ respectively $h_R = 1.2\text{m}$ above the ground with a distance $d = 15\text{m}$.

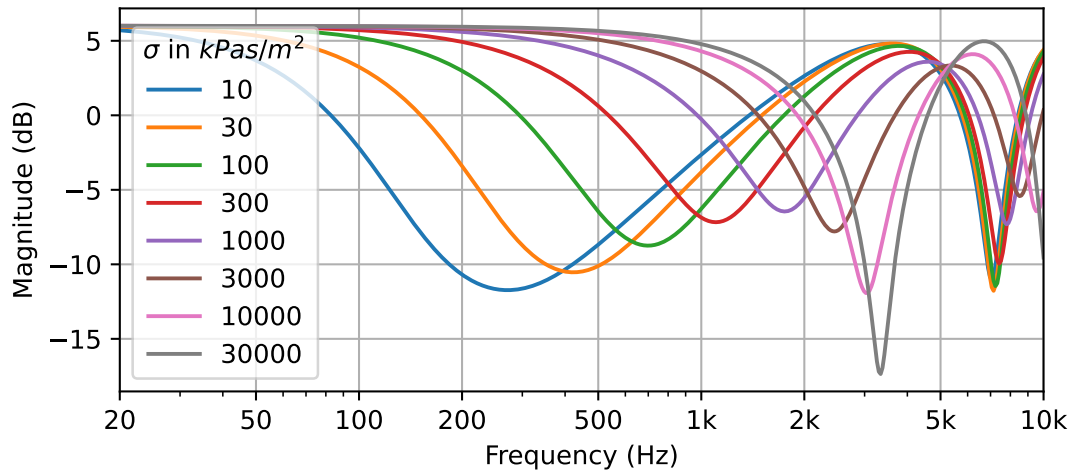


Figure 5.1: Relative sound pressure level for source height $h_S = 0.3\text{m}$, receiver height $h_R = 1.2\text{m}$ and a separation of $d = 15\text{m}$ with ground characterization by effective flow resistivity as the parameter, cf. Delany and Bazley, Eq. 4.12.

For nearly rigid grounds like asphalt with a high flow resistivity the interference minima are sharp and occur when the direct and reflected paths differ by $1/2$ or $3/2$ wavelength, here at frequencies of about 3.2 and 9.6 kHz which indicates only little phase change on

the reflection. This case asymptotically approaches the plane wave reflection ($\mathcal{Q} \xrightarrow{\sigma \uparrow} \underline{R}$), see also Figure A.3 in Appendix A.4. For softer grounds, there is a progressively increasing phase change upon reflection approaching 180° at high frequencies, see Figure A.4 in Appendix A.4. Thus, the interference minimum occurs when the path length difference is one wavelength, which here corresponds to a frequency of 6.4 kHz. The curves exhibit a broad and relatively shallow low-frequency minimum, ranging from approximately 200 to 2000 Hz, depending on the ground's effective flow resistivity, which matches the observations of the measurements presented in Chapter 3.

Figure 5.2 shows the polar distribution of the total normalized sound pressure level of a source located $h_S = 0.1\text{m}$ above the ground sampled at a receiver circle of 1m radius around the source for a grassland ground modeled with $\sigma = 150\text{kPas/m}^2$.

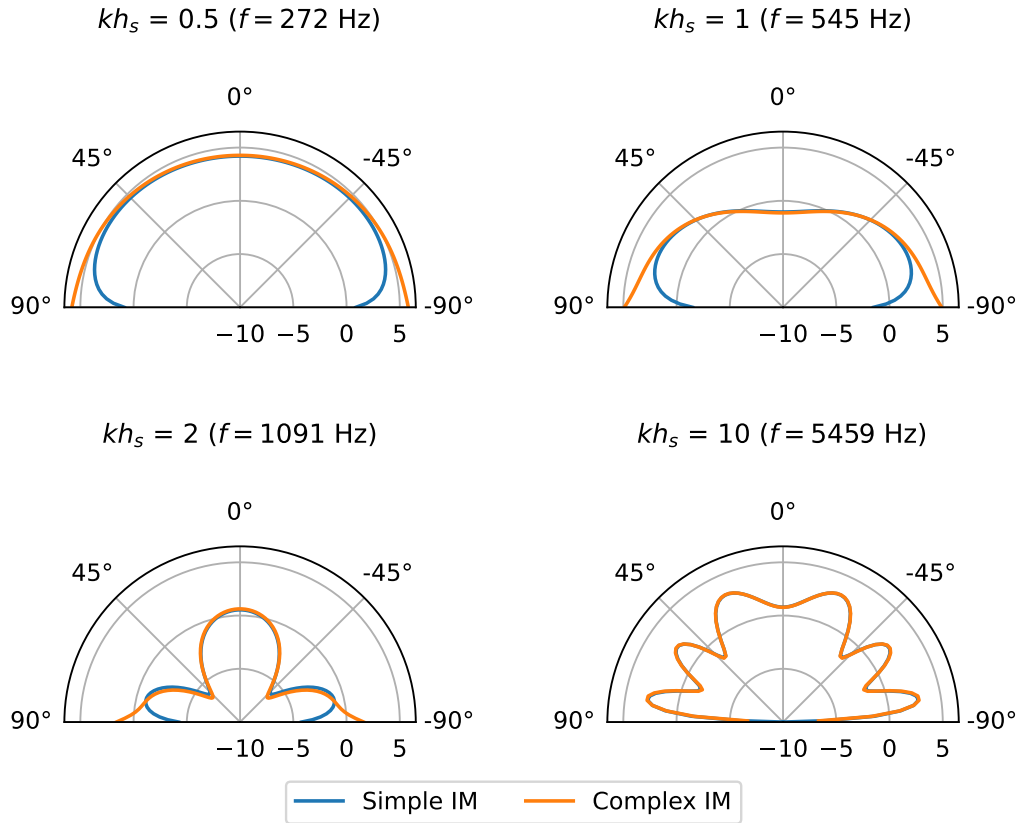


Figure 5.2: Total SPL pattern of direct and reflected wave in dB of a source located in $h_S = 0.1\text{m}$ above a ground for different Helmholtz numbers. The SPL is normalized to the source's SPL in 1m distance. The ground is characterized by an effective flow resistivity of $\sigma = 150\text{kPas/m}^2$ (grassland).

The polar plot shows the expected and nearly omnidirectional reflection pattern for low Helmholtz numbers $kh_S < 1$ and side lobe formation for higher ones for both the simple and complex image method (IM). As discussed in Section 4.1, cf. Eq. 4.9, the complex IM approaches the simple IM for high kh_S . However, for low Helmholtz numbers the phenomenon of the ground wave, which was observed for low frequencies and at low grazing angles, introduced in Section 3.3 in detail, is nicely visible and shows a significant difference to the simple IM.

The simple IM introduced in Section 2.3 and Eq. 2.6 uses a complex frequency and angle dependent reflection factor $\underline{R}(f, \theta)$, cf. Eq. 2.4, and the complex IM contains the complex frequency and angle dependent spherical reflection factor $\underline{Q}$, cf. Eqs. 4.10 and 4.11. However, a further simplified version of the simple IM is used in the *Image Edge Model* (IEM) [13] of the auralization framework *Virtual Acoustics* [14] and just accounts for the absolute value of the plane wave reflection factor $|\underline{R}(f, \theta = 45^\circ)|$ at a fixed angle of $\theta = 45^\circ$ in third-octave bands in Eq. 2.6. This method is referred to in the following as $|R_{45^\circ}|$ -IM. The impedances are obtained with the Delany and Bazley Model (Eq. 4.12) in all methods. Figure 5.3 shows the differences between the presented methods.

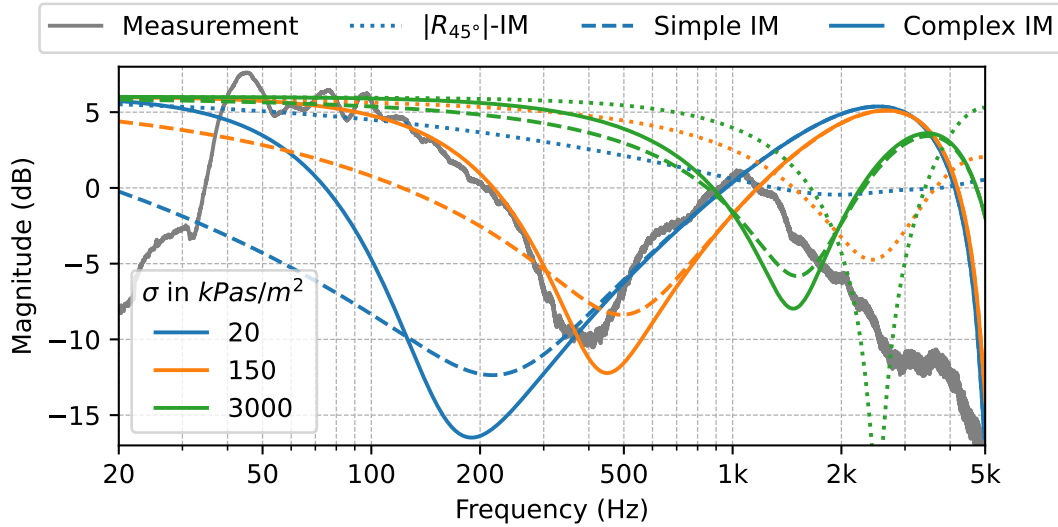


Figure 5.3: Relative SPL predicted by $|R_{45^\circ}|$ -, simple, and complex IM and a measurement from the IHTA-Park (grassland). Setup for both simulation and measurement: source height $h_S = 1.45\text{m}$, receiver height $h_R = 1.2\text{m}$, separation $d = 51\text{m}$. $\sigma = 20\text{kPas/m}^2$ corresponds to snow, $\sigma = 150\text{kPas/m}^2$ to grassland as present in the IHTA-Park, cf. Table A.1.

The $|R_{45^\circ}|$ -IM differs greatly from the other methods in all cases as its first interference minimum occurs at a much higher frequency, which can be calculated by Eq. 3.1 and this is due to the fact that no phase information is included, cf. Figure A.4. The frequency response is just weighted by a frequency dependent real positive number between 0 and 1. The harder the ground, the deeper is dip in the frequency response, as Figure 5.3 shows and one would expect. In contrast, the simple and complex IM model the expected phenomena better and are more similar to each other. Both contain phase information and therefore the interference minima are shifted to lower frequencies and become wider as the effective flow resistivity decreases, i.e. the material becomes more porous. However, as already discussed in Figure 5.2, especially for low frequencies or small grazing angles, let's say small Helmholtz numbers, and the lower the effective flow resistivity, the difference between the simple and complex IM is more significant and is up to 7dB in the simulated scenarios depicted in Figure 5.3. The higher sound pressure level for frequencies below the first interference minimum can be explained by the presence of the ground wave, which occur at very low frequencies, see Section 3.3. In addition, the complex IM models a slightly deeper first interference minimum than the simple IM, which was also already visible in Figure 5.2. Finally, the complex IM clearly aligns best with the measurement when using an effective flow resistivity assumed to be $\sigma = 150\text{kPas/m}^2$, which is according to Table A.1 at the lower end for values used for grassland, but still an appropriate choice as the grass is high and the ground is not too solid in the IHTA-Park.

In summary, it can be said that the complex IM models the observed and discussed phenomena and also matches the measurements quite well, even with a very simple model for the impedance. However, the simple IM, which accounts for a frequency and angle dependent reflection factor containing a complex phase yields good results as well and similar ones in comparison to the complex IM. As the effective flow resistivity, angle, frequency, i.e. Helmholtz number, are increasing the complex IM approaches the simple IM asymptotically. However, especially for all or some of the mentioned factors low, as it applies to many outdoor scenarios there is a significant difference justifying the complex IM approach supported by matching real-world measurements.

6 Summary and Outlook

This work presented an analysis of sound propagation in outdoor environments, focusing on the ground reflection and special effects due to the ground. The *Complex Image Method* was presented from literature to model sound as spherical waves, providing a more accurate representation of observed real-world phenomena compared to the *Simple Image Method* for shallow grazing incidence. The importance of considering ground impedance or rather estimating it by the effective flow resistivity when modeling sound propagation and the need for a complex reflection including phase information was highlighted and discussed. Case studies and a comparison of state-of-the-art methods and the proposed method against real-world measurements successfully demonstrated that the *Complex Image Method* provides better alignment with observed phenomena and measurements.

Future research could investigate methods for integrating these effects into auralization models, such as the Image Edge Model [13, 14], which currently do not properly account for ground effects. This could also include the impedance models presented in Section 4.2, which were not further investigated in the simulations in this work. Another problem encountered during the work was that there is little data available for the validation of the methods in the case of shallow grazing angles. Impedance values are mostly determined in impedance tubes, and thus, for normal incidence, and direct measurement is very difficult. Conducting outdoor measurements and developing methods that enable impedance measurements at small grazing angle is necessary and could help to better distinguish between distinct ground effects and improve predictions of outdoor sound propagation at very shallow grazing incidence.

A Appendix

A.1 IHTA-Park Measurement Setup

The park around the Institute for Hearing Technology and Acoustics at RWTH Aachen University, Kopernikusstraße 5, 52074 Aachen is called IHTA-Park.

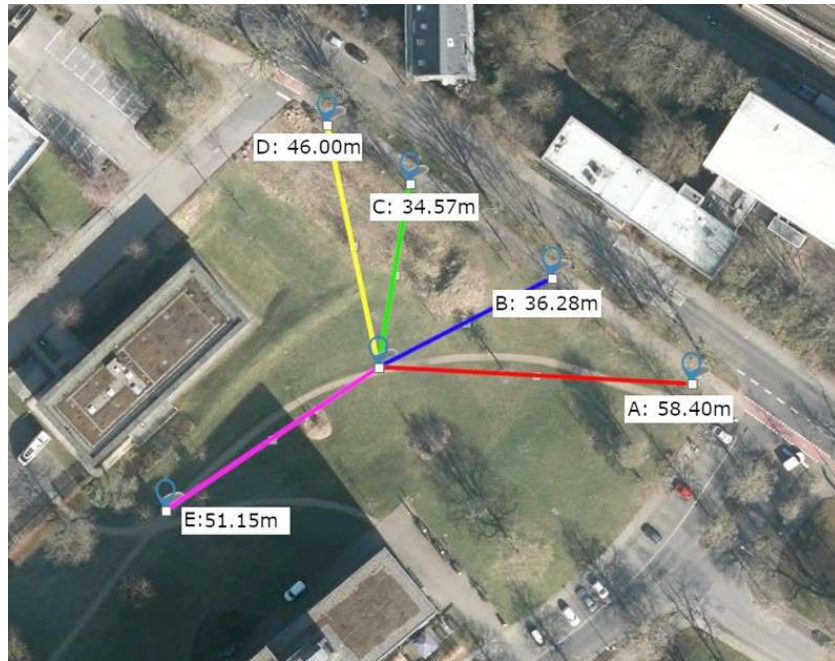


Figure A.1: IHTA-Park measurement setup. Source was placed at a height of $h_S = 1.45\text{m}$ above the ground, receiver at $h_R = 1.2\text{m}$.

A.2 Derivation of The Complex Image Method

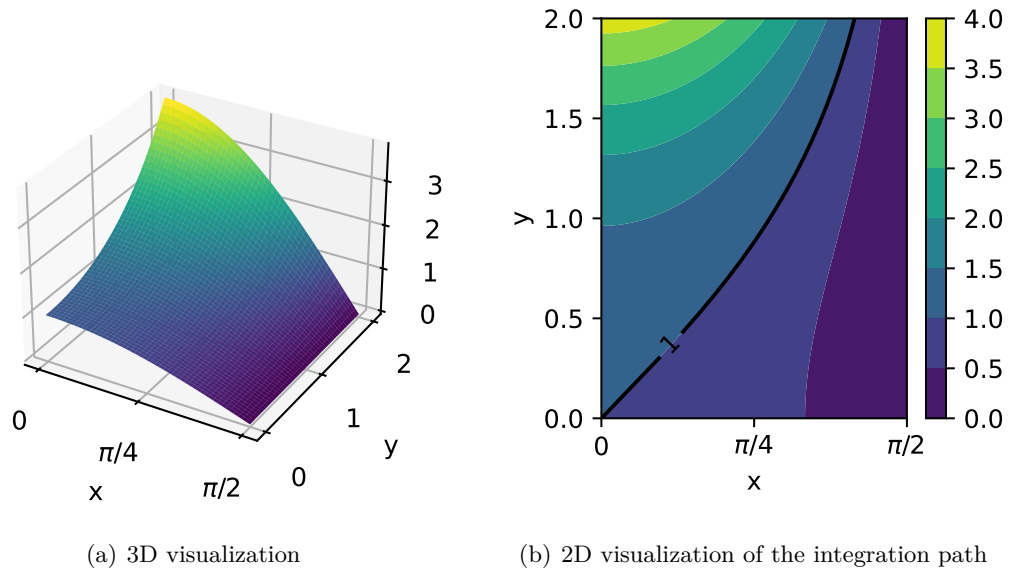


Figure A.2: Real part of the complex cosine $\cos(x + jy)$

A.3 Ground Characterization by Effective Flow Resistivity

Description of surface	Effective flow resistivity kPas/m ²
0.1 – m new fallen snow, over older snow	10 – 30
sugar snow	25 – 50
floor of evergreen forest	20 – 80
airport grass or old pasture	150 – 300
roadside dirt, ill-defined, small rocks up to 0.01m mesh	300 – 800
sandy silt, hard packed by vehicles	800 – 2500
thick layer of clean limestone chips, 0.01 to 0.025m mesh	1500 – 4000
old dirt roadway, small stones with interstices filled by dust	2000 – 4000
earth, exposed and rain-packed	4000 – 8000
very fine quarry dust, hard packed by vehicles	5000 – 20000
asphalt, sealed by dust and use	~ 30000
upper limit, set by thermal conduction and viscosity	2×10^5 to 1×10^6

Table A.1: Characterization of various ground surfaces. The values of effective flow resistivity were obtained by matching transmission spectra measured in situ with spectra predicted theoretically. Tab. taken from [6].

A.4 Properties of the compared Reflection Factors

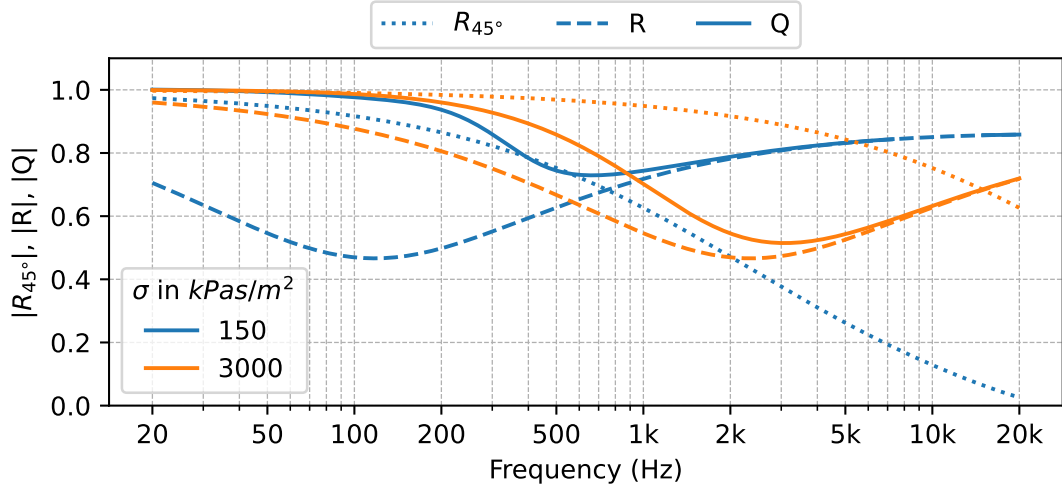


Figure A.3: Absolute Value of R_{45° , $\underline{R}$, and $\underline{Q}$ for different effective flow resistivities σ using Delany and Bazley Model (eq. 4.12) for the impedance. The last two are calculated for source height $h_S = 1.45\text{m}$, receiver height $h_R = 1.2\text{m}$, separation $d = 51\text{m}$.

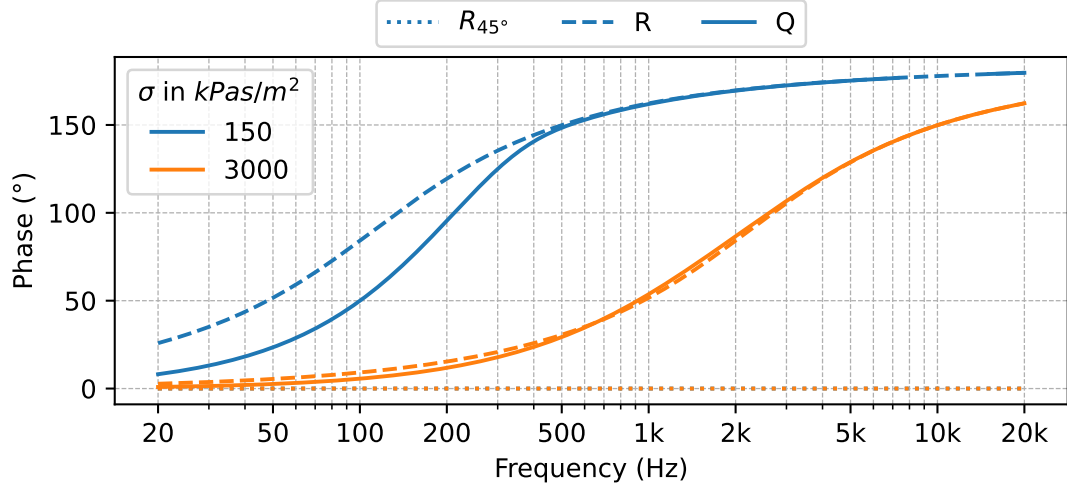


Figure A.4: Phase of R_{45° , $\underline{R}$, and $\underline{Q}$ for different effective flow resistivities σ using Delany and Bazley Model (eq. 4.12) for the impedance. The last two are calculated for source height $h_S = 1.45\text{m}$, receiver height $h_R = 1.2\text{m}$, separation $d = 51\text{m}$.

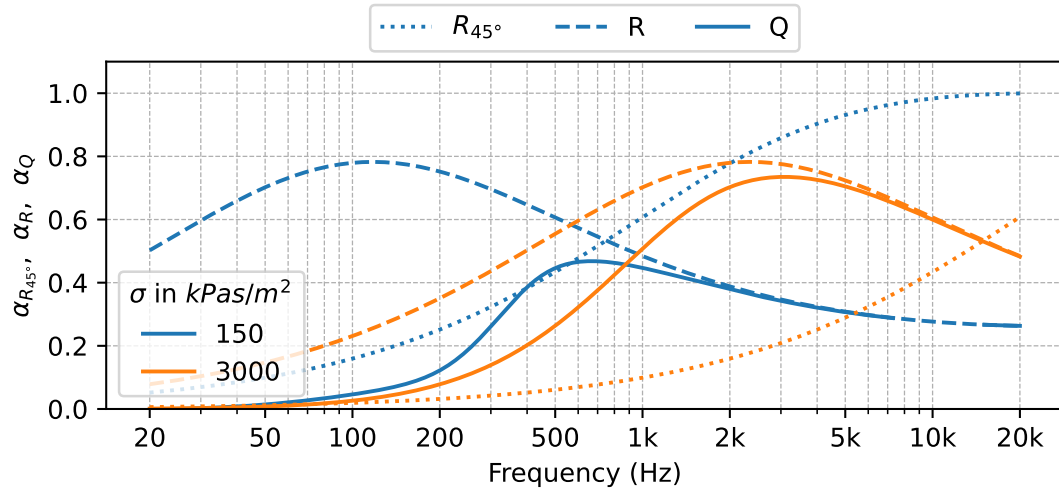


Figure A.5: Absorption coefficient α , cf. 2.5 for R_{45° , R , and Q for different effective flow resistivities σ using Delany and Bazley Model (eq. 4.12) for the impedance. The last two are calculated for source height $h_S = 1.45\text{m}$, receiver height $h_R = 1.2\text{m}$, separation $d = 51\text{m}$.

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